End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name:<B.Tech. > Semester:<1st >
(For 2025 admitted students)
December, 2025

Subject Code: BAS 101	Subject: Applied Mathematics
Time :3 Hours	Maximum Marks :60

Note : Q1 is compulsory.
Attempt any one question from the Units I, II, III & IV.
Calculators are NOT allowed.

Q1		(2.5*8=20)	CO Mapping
	(a) Prove that the eigenvalues of the unitary matrix are of unit modulus.		CO1
	(b) Two eigenvalues of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are equal to one each, find the eigenvalues of A^{-1}.		CO1
	(c) Find the nth derivative of $\frac{1}{6x^2-5x+1}$.		CO2
	(d) Test the function $f(x,y) = \begin{cases} \frac{x^3-y^3}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$ for continuity.		CO2
	(e) Evaluate $\int_0^4 \int_{x=0}^{2\sqrt{z}} \int_0^{\sqrt{4z-x^2}} dxdydz$.		CO3
	(f) Compute the $Jacobian$ if $x = u(1-v), y = uv$.		CO3
	(g) Find the unit normal to the surface $x^2y + 2xz^2 = 8$ at the point $(1,0,2)$.		CO4
	(h) Prove that $div\left(\emptyset\vec{A}\right) = (grad\ \emptyset).\vec{A} + \emptyset.\ div\vec{A}$.		CO4
	UNIT-I		**CO Mapping**
Q2	(a) Solve the homogeneous system $3x + 4y - z - 6w = 0$ $2x + 3y + 2z - 3w = 0$ $2x + y - 14z - 9w = 0$ $x + 3y + 13z + 3w = 0.$ (b) For the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, find the modal matrix P.	(5*2=10)	CO1
	OR		
Q3	(a) Using Gauss-Jordan method, find inverse of a matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & 4 \end{bmatrix}$.	(5*2=10)	CO1

	(b) For the matrix $A = \begin{bmatrix} 7 & -2 & 1 \\ -2 & 10 & -2 \\ 1 & -2 & 7 \end{bmatrix}$, find the modal matrix P.		
	UNIT-II		**CO Mapping**
Q4	(a) Expand $\tan^{-1}\left(\frac{y}{x}\right)$ in Taylor series at $(1,1)$ up to second degree terms. (b) Test the function $f(x,y) = x^3y^2(6-x-y)$ for maxima and minima and hence, determine the maximum/minimum value of $f(x,y)$.	**(5*2=10)**	CO2
	OR		
Q5	(a) If $x = rcos\theta$ and y= $rsin\theta$, prove that $(x^2-y^2)\left(\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2}\right) + 4xy\frac{\partial^2 u}{\partial x \partial y} = r^2\frac{\partial^2 u}{\partial r^2} - r\frac{\partial u}{\partial r} - \frac{\partial^2 u}{\partial \theta^2}$, where u is any twice differentiable function of x and y. (b) Prove that if parameter of a triangle is constant, then its area is maximum if the triangle is equilateral.	**(5*2=10)**	CO2
	UNIT-III		**CO Mapping**
Q6	(a) Trace the curve $a^2y^2 = x^2(a^2-x^2)$. (b) Change the order of integration in $\int_0^1 \int_y^{2-y} xydxdy$ and evaluate the integral.	**(5*2=10)**	CO3
	OR		
Q7	(a) Find the area enclosed by the curves $y = 2-x$ and $y^2 = 2(2-x)$ using double integration. (b) Transform to polar coordinates and evaluate the integral $\int_0^2 \int_0^{\sqrt{4-x^2}} (x^2y + y^3)dxdy$.	**(5*2=10)**	CO3
	UNIT-IV		**CO Mapping**
Q8	(a) State Green's theorem and use it to evaluate $\oint_C [(x^2 - y^2)dx + 2xydy]$, where C is the closed curve of the region bounded by $y^2 = x$ and $x^2 = y$. (b) Find the scalar potential and work done by applying the force $\vec{F} = (2xy + z^3)\hat{\imath} + x^2\hat{\jmath} + 3xz^2\hat{k}$ in moving an object from (1, -2, 1) to (3, 1, 4).	**(5*2=10)**	CO4
	OR		
Q9	(a) Verify Gauss divergence theorem for $\vec{F} == (x^2 - yz)\hat{\imath} + (y^2 - zx)\hat{\jmath} + (z^2 - xy)\hat{k}$ taken over the rectangle parallelepiped $0 \le x \le a,\ 0 \le y \le b,\ 0 \le z \le c$. (b) Explain the physical interpretation of curl. Also, find a, b, c If $(x + y + az)\vec{\imath} + (bx + 2y - z)\vec{\jmath} + (-x + cy + 2z)\vec{k}$ is Irrotational.	**(5*2=10)**	CO4